

AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous by default. [Supervisable by design.](#)

ROBUST

SUPERVISABLE

BROAD SEARCH

The run, in four moves

A single route from the problem to the proof.

01



PROBLEM

Why current agents fail

Set the task and its failure modes.

02



SYSTEM

A system for sustained research

Keep experience while resetting the scientist.

03



USER EXPERIENCE

Run the research journey

Move from task contract to retained state.

04



EVALUATION

From claim to evidence

Measure what survives the run.

Automating ML experimentation

Given a dataset and a scoring metric, an agent must build a model and improve it through repeated experiments.

01	02	03	04	05
Read the problem	Inspect data	Engineer features	Train + tune	Evaluate
What should the model predict?	What patterns does the data contain?	Turn raw data into useful model inputs	Fit a model and adjust its settings	Check performance on validation data
REFLECT + REVISE				
The agent uses the results to choose what to change, revises the code, and runs the next experiment.				

THE AGENT'S JOB

Write the code and run the full loop autonomously.

This includes building the pipeline and deciding which experiments to try next.

THE CHALLENGE GOAL

Reproduce, then improve on the organizer's baseline.

Develop with training data and public validation. The organizer scores the final submission on a hidden test set.

Why current agents fail

01 WORKFLOW



Observe → Code → **Error** → **Recover**

Coding
Harness

Runtime
Tools

Acquire missing tools at runtime.

- ✓ Resume after unexpected errors.
- ✓ Extend with Skills & MCP.

02 SINGLE AGENT



last frame — next move

history-shaped

single path

Fresh reasoning. Audited evidence.

- ✓ Reopen search.
- ✓ Check the science.

03 TREE SEARCH



READ ONCE



BRANCH EDITS

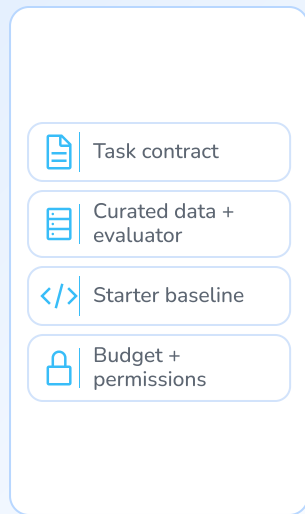
Persist the read. Vary the hypothesis.

Reuse context across experiments.

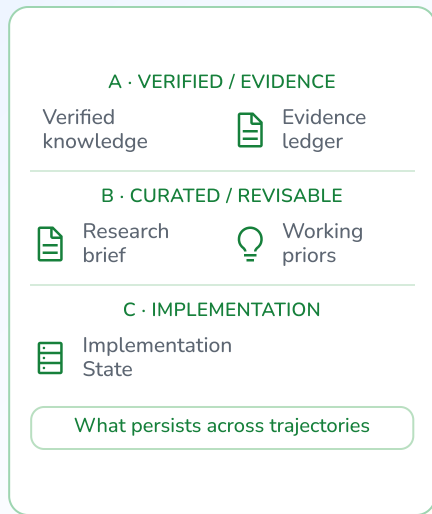
- ✓ Raise task granularity.
- ✓ Parallelize across branches.

A system for sustained research

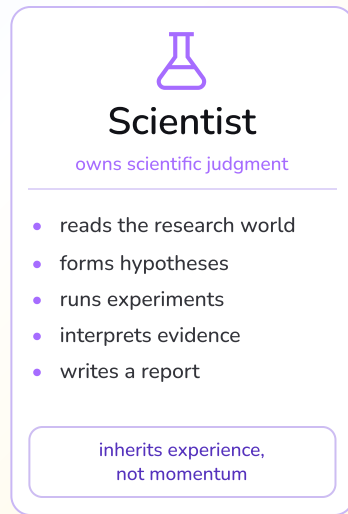
Environment



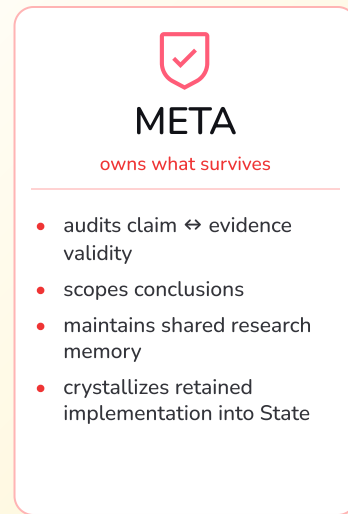
Research World



Fresh Scientist



META Review



Selective persistence / trajectory handoff

Coding-agent harness + runtime



SciOdyssey

A research layer over your agent harness.

We are not tokenmaxxing

PROJECT-LEVEL RESOURCE ACCOUNTING

~\$10

subscription-equivalent usage from first line of code to final result

coding · debugging · agent runs · all experiments

MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

COMPUTE

1 × MacBook M2

Development and experiments ran on one consumer laptop.

Retained-run telemetry

48.240M

total tokens

including cache-read input

4.020M

non-cache

input + output

0

GPU-hours

CPU / NumPy training

Model diversity improves the result.

Cycling GPT and Gemini brings different search styles into one research world.

Deepen an idea

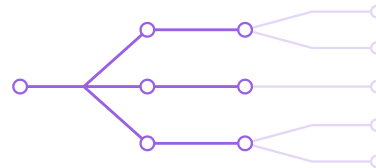
GPT 5.6 Luna



Refine the current approach.
Follow promising details further.

Open new directions

Gemini 3.7 Flash



Try alternative approaches.
Challenge the current direction.

Model diversity improved the retained result.

GPT → **Gemini** → **GPT** → **Gemini**

🔄 Shared research memory · Gemini META

Public-validation Primary

Gemini-only

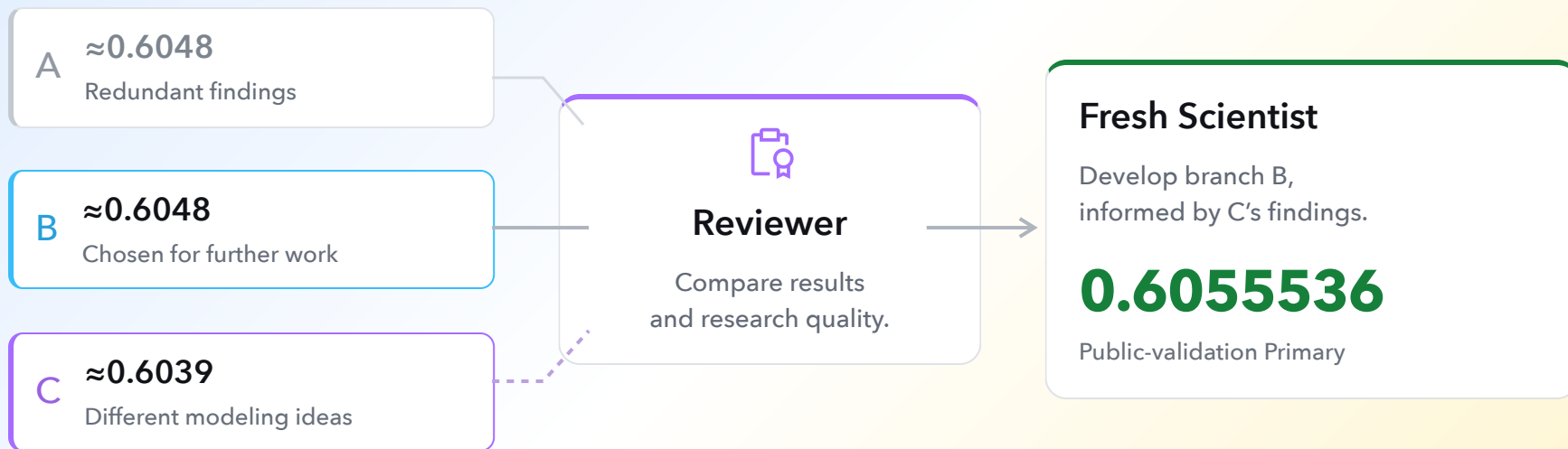
GPT + Gemini

0.6052

0.6059363

Explore in parallel. Build on the findings.

One branch continues. Another contributes a useful perspective.



A lower-scoring branch can still contribute an idea.

The new model used pairwise ranking, higher rank and rank-normalized ensembling.

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS



Give every agent a branch.

main

clean trunk

● agent A / branch

● agent B / branch

● agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP



Leave the next move in GitHub.

01

Issue /
prompt

inject the task



02

GitHub
Action

run the agent



03

Commit

review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

Research is an **open-world** task.
We need an **open-world** coding agent.

01 OPEN ACTION SPACE



Shell



Files



Browser



Git

Use what exists

02 SELF-RECOVERY



Read errors



Inspect docs



Inspect
source



Install
packages

Fix what breaks

03 RUNTIME EXTENSION



MCP



Skills



CLI



Packages

Add what's missing

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD



Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture ·
integration



Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles ·
evidence review



Shilin Xu

DATA / METRICS

Data workflow · contracts
· alignment



GE GAO

RUNTIME / RELIABILITY

Execution · testing ·
integration support



Jiran Li

EVIDENCE /
REPRODUCIBILITY

Telemetry · documentation
· packaging